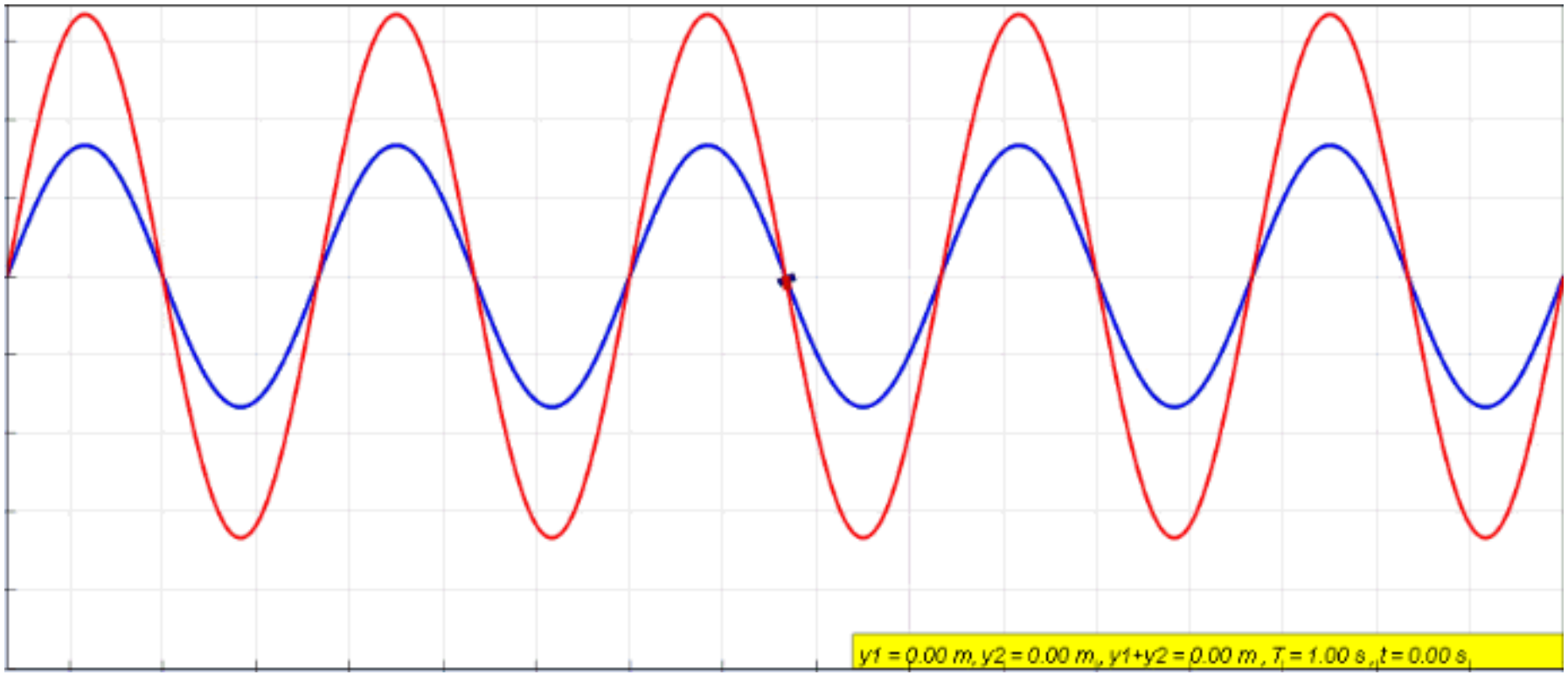
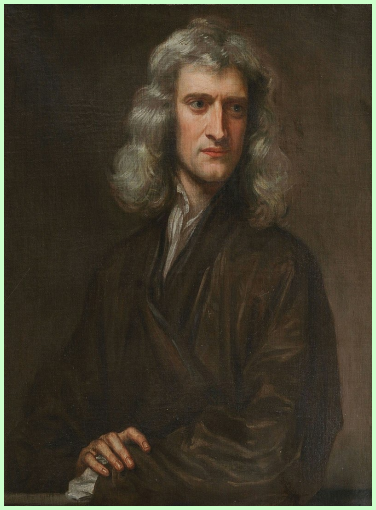
Superposition



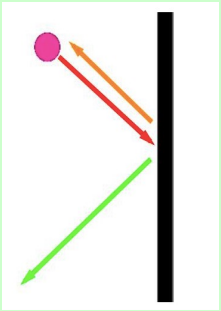
What is light?

"Light is a particle"
(Crepuscule")



Isaac Newton

Light bounces off walls
just like a particle would



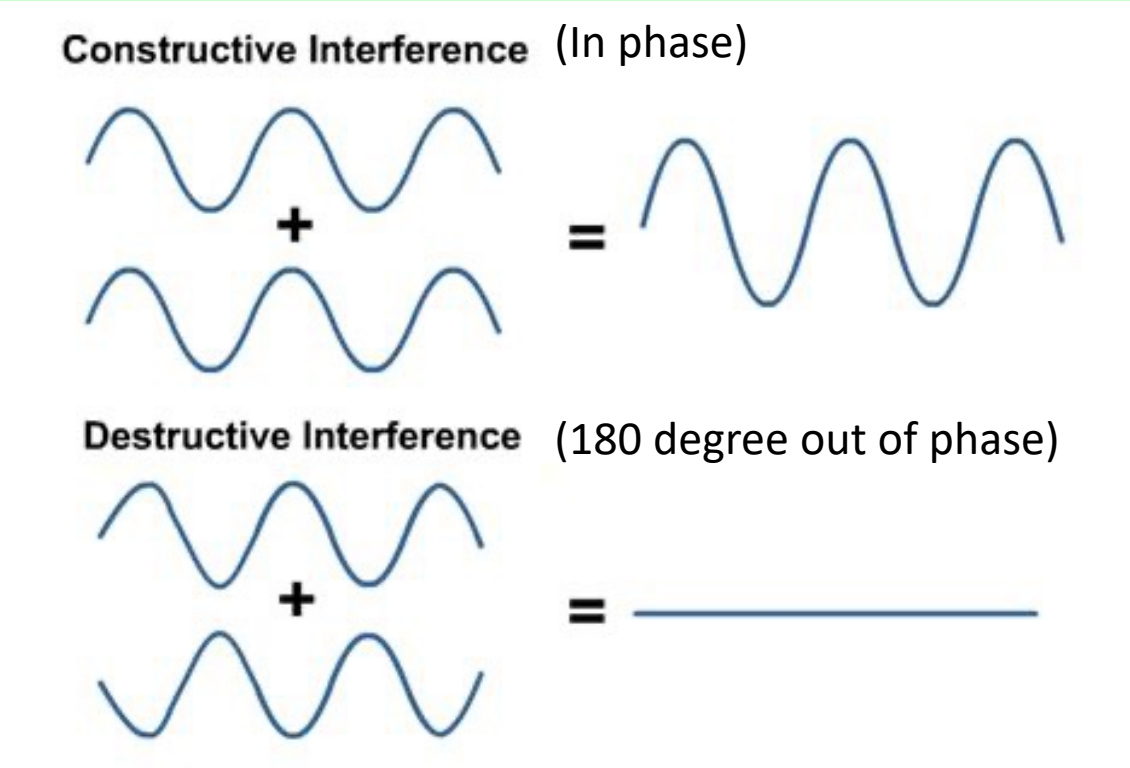
"Light is a Wave"



Christian Huygens

Light follows superposition

Superposition



If two waves are in the same place at the same time: you have to add them:

$$D_{\text{total}}(x,t) = \sum D_i(x,t)$$

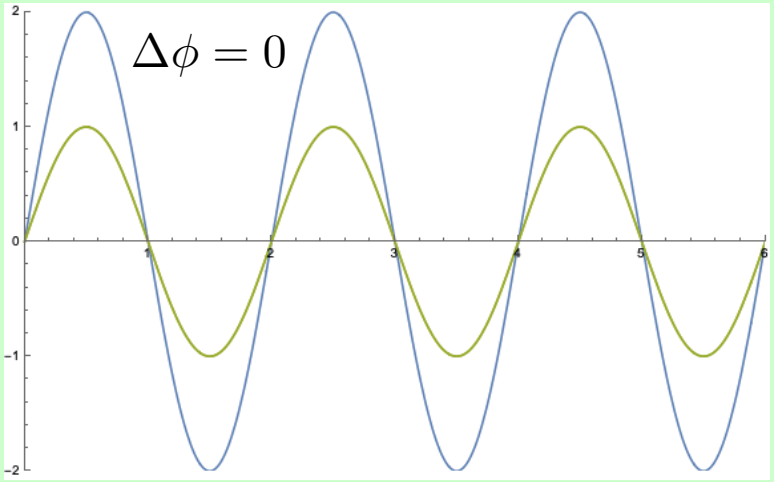
Ex. What happens if you add: (all amplitude =1)

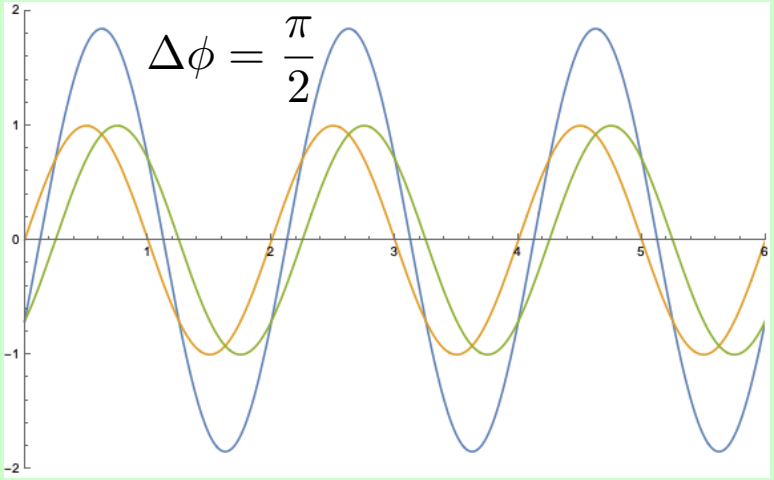
A.) Two waves of wavelength = 2m, in phase

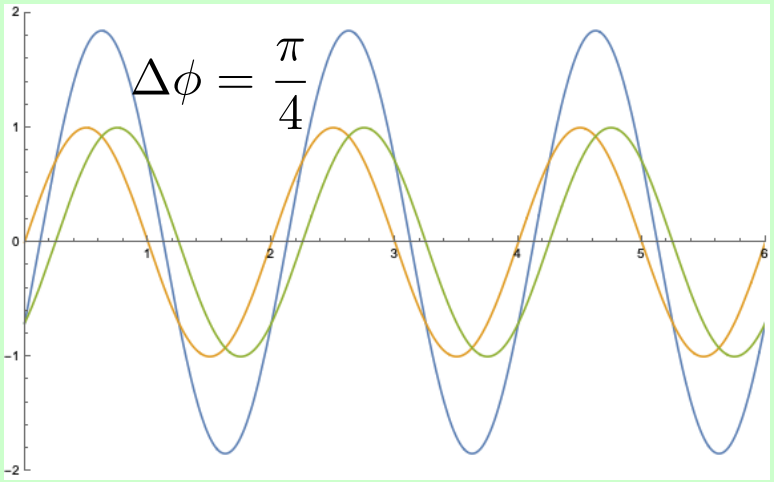
B.) Two waves of wavelength = 2m, 45 degrees out of phase?

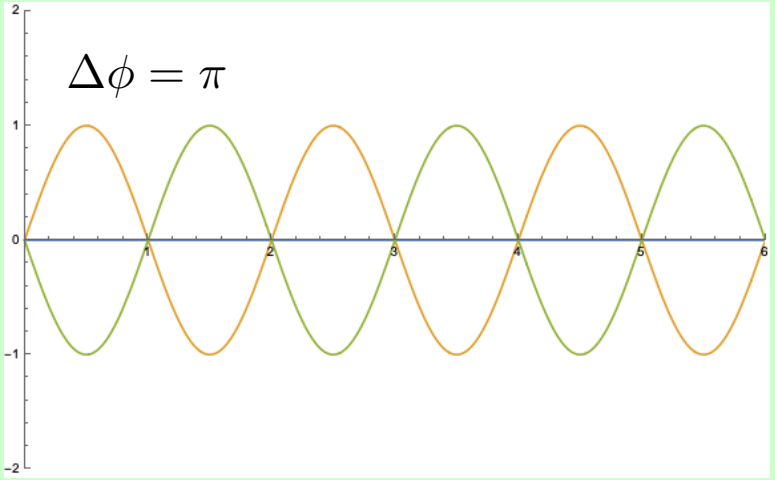
C.) Two waves of wavelength = 2m, 90 degrees out of phase?

D.) Two waves of wavelength = 2m, 180 degrees out of phase



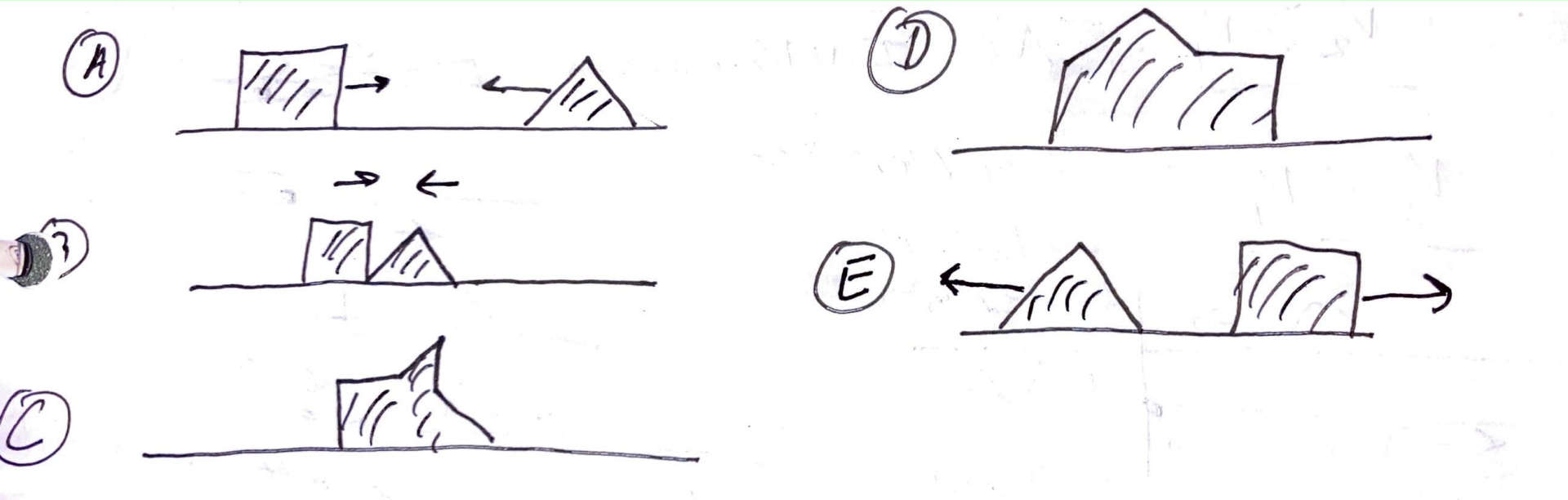




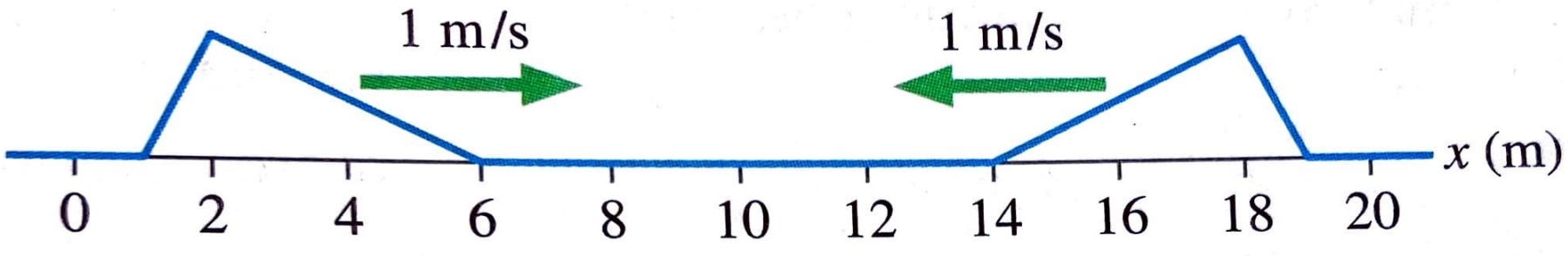


It may be obvious, but take superposition literally!!

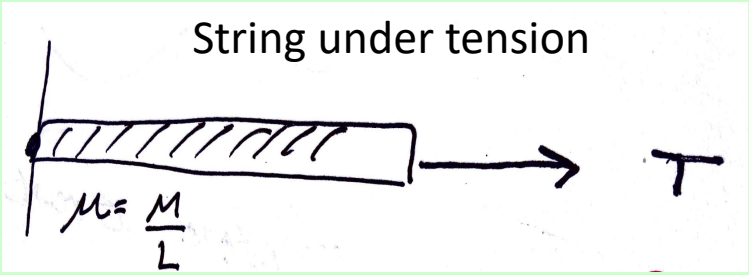
Ex. Two weird pulses passing each other



Ex. What is the shape of the pulse at t=6 seconds?



What sets the travel speed on a string?



$$v = \sqrt{\frac{T}{\mu}}$$

1. More T -> faster
2. Lower μ _> also faster
   (less mass means same force accelerates more)

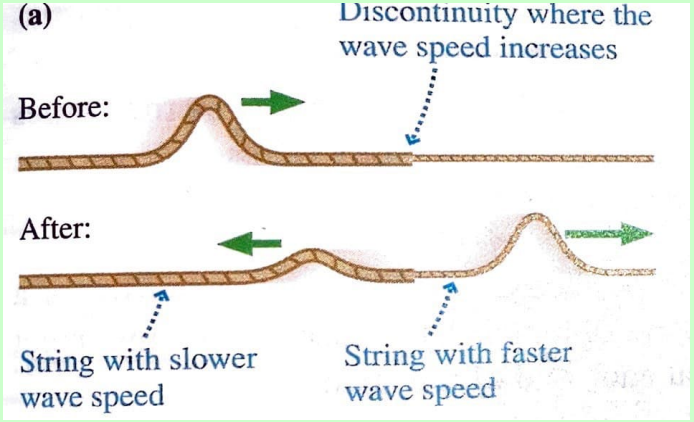
Demo: Slinky

Demo: Shively wave machine

Reflection of pulse at a discontinuity

Wave reaches point where. μ decreases

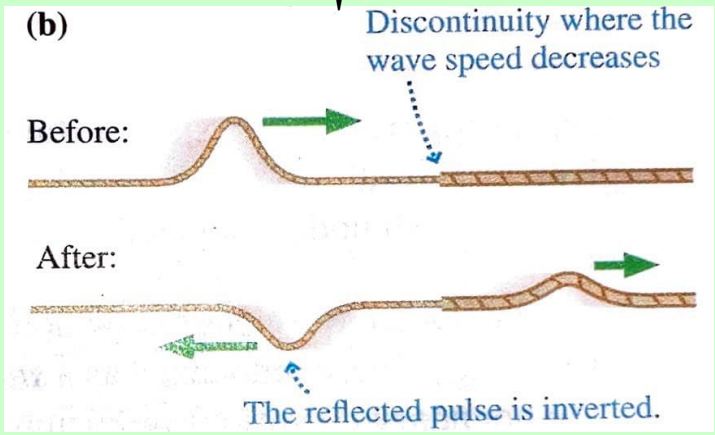
Wave speed, $v = \sqrt{\frac{T}{\mu}}$ increases



Some wave reflected, some transmitted
Because v2> v1, reflected wave is not inverted

Wave reaches point where. μ increases

Wave speed, $v = \sqrt{\frac{T}{\mu}}$ decreases



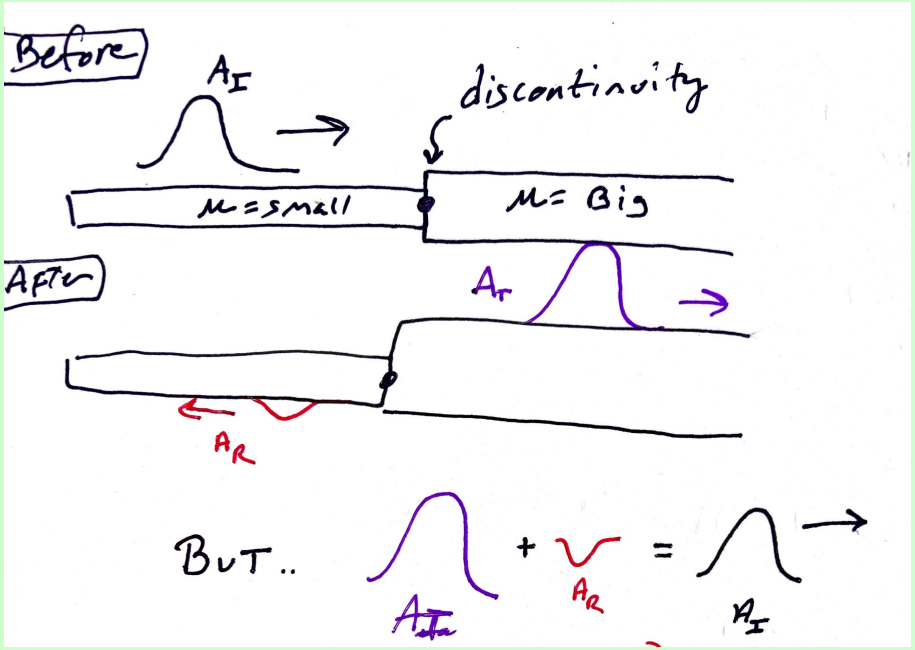
Some wave reflected, some transmitted
Because v2< v1, reflected wave is inverted

If velocity decreases the reflected wave inverts
(or) If density increases, the reflected wave inverts

Incident, Reflected, and Transmitted Amplitudes

You can think of the initial (incident) wavepulse as being conserved through a discontinuity. So the incident Amplitude equals the Reflected plus transmitted amplitudes. A_R counts as negative since the wave is going in opp direction

$$A_I = A_T - A_R$$



Ex. A_I = 1.5 m, A_R = -0.4 m
A_T ?

Think of shoving wave up on costal shelf, Most of wave height is transmitted, but the reflected portion is shifted by 180 deg

Wave Impendence

How do you determine the Reflected and Transmitted amplitude if you know the Incident amplitude.

What is conserved is really the **momentum, p** of the wave. But when solving the wave Equation on a string its easier to work **of Impedance, Z**

$$p = mv$$

(momentum)

$$z = \rho v$$

(Impedance)

For the case of a string, the density is really the linear mass density, μ

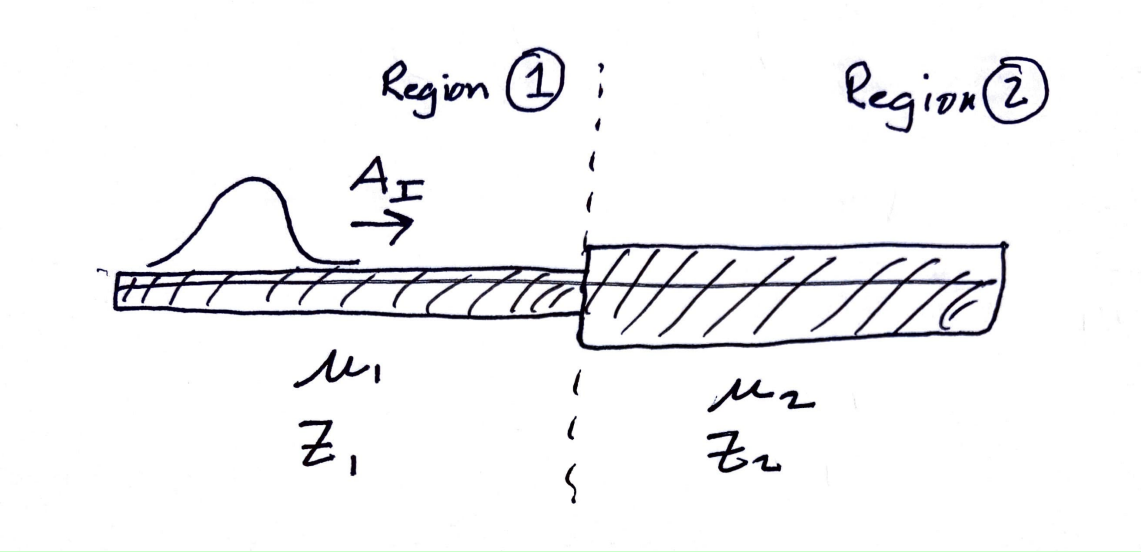
$$z = \rho v = \mu v = \mu\sqrt{\frac{T}{\mu}} = \boxed{\sqrt{\mu T}}$$

Ex. What is the impedance of a wave on a string with a tension of 100 N and a linear mass density of 0.02 kg/m?

Since the momentum (impedance) is conserved, its possible to determine
The reflected and transmitted amplitudes

$$A_R = \frac{Z_I - Z_2}{Z_1 + Z_2} A_I$$

$$A_R = \frac{2Z_I}{Z_1 + Z_2} A_I$$



So for each region, compute Z then use the formulae to find A_R and A_T

EX. A string has a discontinuity where the linear mass density changes from 1 g/m to 8 g/m. The string is under constant tension of T = 80 N.

A. Compute the velocity of the waves in each region
B. Determine the Impedance of the wave in each region
C. Compute the reflection and transmission amplitudes of the wave
D. Show that $A_I = A_T - A_R$
E. Draw a picture of this scenario after the pulse has hit the discontinuity

Ex. A string has a discontinuity where the linear mass density changes from 10 g/m to 5 g/m. The string is under constant tension of T = 10 N.

A. Compute the velocity of the waves in each region
B. Determine the Impedance of the wave in each region
C. Compute the reflection and transmission amplitudes of the wave
D. Show that $A_I = A_T - A_R$
E. Draw a picture of this scenario after the pulse has hit the discontinuity